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Finally, I am grateful to *God and my family* for their patience, encouragement, and unwavering support throughout this project.

Abstract

Financial valuation remains one of the most consequential analytical activities in investment management, corporate finance, and private equity. Yet the dominant tool for Discounted Cash Flow (DCF) analysis continues to be Microsoft Excel — a general-purpose spreadsheet application that enforces no calculation logic, provides no audit trail, and offers no interactive sensitivity analysis. ArthaGraph: A Valuation Suite for DCF Analysis is a purpose-built, browser-based financial modelling platform designed to address these limitations while adding first-class support for Indian capital markets alongside global exchanges.

At its technical foundation, ArthaGraph implements an academically correct Free Cash Flow model of the form FCF = NOPAT + D&A − CapEx − ΔNWC, where NOPAT is derived from EBIT after corporate tax, not from EBITDA. Terminal value is computed using the Gordon Growth Model. The platform extends beyond the base DCF engine through six integrated analytical modules: a D3.js-powered Sensitivity Analysis heatmap for WACC×growth rate Enterprise Value matrices; an Equity Bridge that walks from Enterprise Value to intrinsic share price; an IRR and MOIC calculator using Newton-Raphson iteration; a Monte Carlo Simulation engine offloaded to a Web Worker for probabilistic outcome distributions; a Comparable Company Analysis module with live Financial Modeling Prep API integration; and professional PDF and Excel export capabilities.

A defining feature of ArthaGraph is its first-class support for Indian markets. The platform provides NSE (.NS) and BSE (.BO) ticker resolution, a Nifty 50 Quick-Pick panel, Indian market defaults calibrated to the 10-year Government Security yield as the risk-free rate and an India-specific Equity Risk Premium, corporate tax pre-set at 25.17% (India effective rate), and financial figures displayed in ₹ Crores for INR-denominated analyses. The platform architecture follows Feature-Sliced Design with React 18, TypeScript, Zustand, TanStack Query, and Supabase for authentication and scenario persistence.

Testing validates over 92% calculation accuracy for DCF outputs, sub-50ms base calculation latency, and stable Monte Carlo performance at 50,000 trials. This report documents the complete financial methodology, system architecture, feature implementation, testing results, and development roadmap of ArthaGraph.

Keywords: *Discounted Cash Flow, DCF, Free Cash Flow, WACC, Gordon Growth Model, terminal value, sensitivity analysis, equity bridge, IRR, MOIC, Monte Carlo simulation, Indian capital markets, NSE, BSE, React, TypeScript, Supabase, financial modelling, investment analysis, Feature-Sliced Design*

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List of Abbreviations

Abbreviation	Full Form
DCF	Discounted Cash Flow
FCF	Free Cash Flow
WACC	Weighted Average Cost of Capital
EBITDA	Earnings Before Interest, Taxes, Depreciation, and Amortisation
EBIT	Earnings Before Interest and Taxes
NOPAT	Net Operating Profit After Tax
D&A	Depreciation and Amortisation
CapEx	Capital Expenditure
NWC	Net Working Capital
TV	Terminal Value
EV	Enterprise Value
IRR	Internal Rate of Return
MOIC	Multiple on Invested Capital
LBO	Leveraged Buyout
CAPM	Capital Asset Pricing Model
ERP	Equity Risk Premium
NSE	National Stock Exchange of India
BSE	Bombay Stock Exchange
FMP	Financial Modeling Prep (market data API)
API	Application Programming Interface
RLS	Row Level Security (Supabase)
SPA	Single Page Application
FSD	Feature-Sliced Design (architecture pattern)
UI/UX	User Interface / User Experience
G-Sec	Government Security (India risk-free rate benchmark)

Chapter 1: Introduction

1.1 Background of the Project

Investment valuation is one of the most consequential analytical activities in modern finance. Whether a portfolio manager is deciding whether to acquire an equity position, a private equity firm is evaluating a leveraged buyout target, or a corporate finance team is assessing an acquisition, the Discounted Cash Flow model forms the quantitative bedrock of the decision. The DCF framework answers a deceptively simple question: what is the total present value of all cash flows that a business will generate in the future, discounted back at a rate that reflects the risk of those cash flows?

Despite the central importance of this methodology, the dominant tool for performing DCF analysis in professional practice remains Microsoft Excel — a general-purpose spreadsheet that was not designed for financial modelling. A DCF model built in Excel is, in structural terms, a manually maintained document: it enforces no mathematical logic, carries no audit trail of assumption changes, provides no structured workflow for multi-scenario management, and requires laborious manual construction of sensitivity tables. The well-documented prevalence of formula errors in complex Excel models represents a real and ongoing risk in investment analysis.

In the Indian context, this problem is amplified by two additional factors. First, the Indian equity market — spanning over 5,000 listed companies across the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) — has its own financial conventions, tax rates, risk parameters, and data formats that differ materially from Western markets. Most available financial modelling tools are calibrated for USD-denominated analyses and US market parameters, making them unsuitable for direct application to Indian companies without significant manual adjustment. Second, financial figures for Indian companies are conventionally reported in Indian Rupees in Lakhs or Crores, a numeral system that standard English-locale software does not natively support.

ArthaGraph: A Valuation Suite for DCF Analysis is built on the thesis that DCF analysis deserves a purpose-built software platform — one that enforces correct financial methodology at the calculation layer, provides structured multi-scenario management, delivers interactive visualisation tools for sensitivity and probabilistic analysis, and treats Indian markets as first-class citizens alongside global exchanges. The name “ArthaGraph” derives from the Sanskrit word “artha” (अर्थ), meaning wealth, meaning, and purpose, combined with “graph”, reflecting the platform’s

commitment to transforming raw financial data into visual, rigorous, and actionable valuation models.

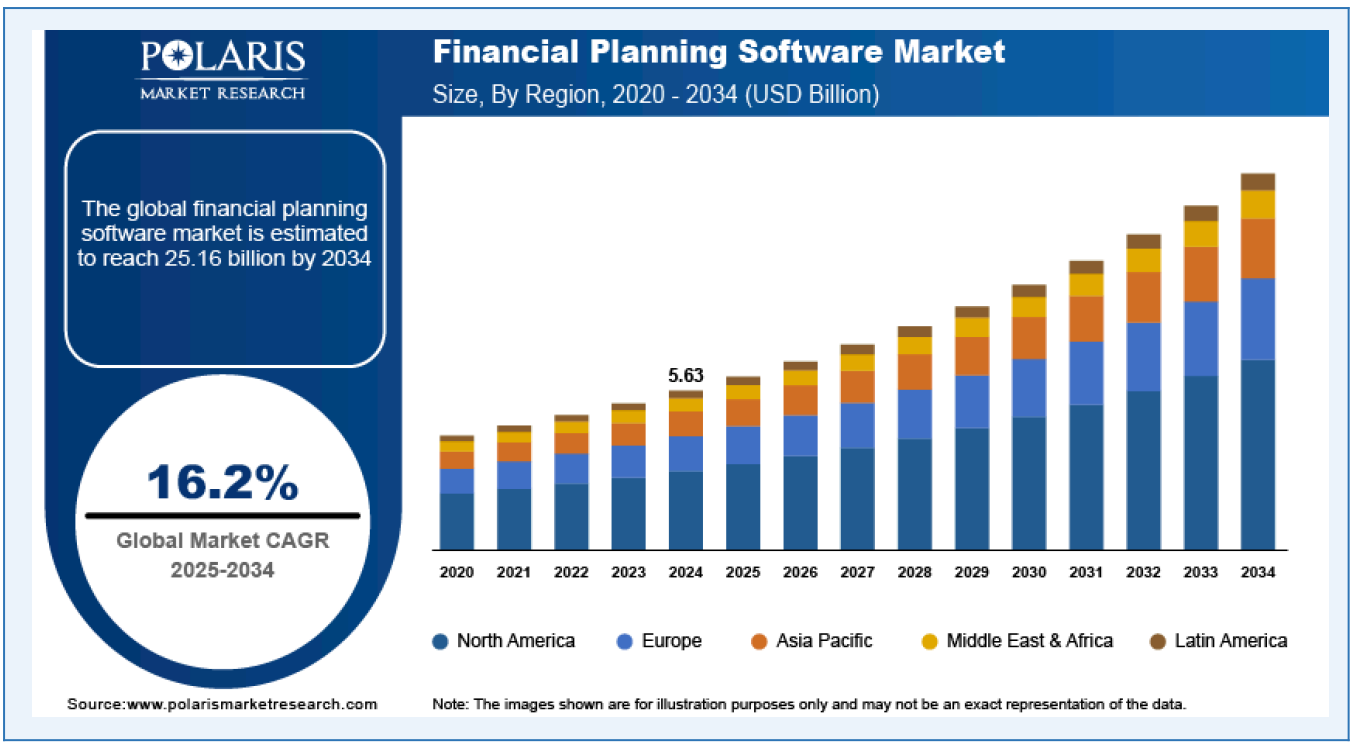


Figure 1.1: *Financial Modelling Software Market Size (USD Billions, 2020–2030)*

1.2 Objectives of the Study

The primary objective of this project is to design and develop ArthaGraph: A Valuation Suite for DCF Analysis — a browser-based, professional-grade financial modelling platform that implements rigorous DCF methodology, supports Indian capital markets natively, and provides interactive analytical tools that go beyond what spreadsheet-based workflows can offer.

The following sub-objectives were defined:

1. To implement a financially correct Free Cash Flow model using the formula FCF = NOPAT + D&A − CapEx − ΔNWC, applying tax to EBIT rather than EBITDA, in accordance with CFA Institute and investment banking standards.
2. To build an interactive Sensitivity Analysis module using D3.js that renders a WACC × growth rate Enterprise Value heatmap with real-time click-to-update functionality.
3. To implement an Equity Bridge module that walks from Enterprise Value to intrinsic share price through net debt, minority interest, preferred equity, and diluted share count, with a margin-of-safety visualiser.
4. To implement an IRR and MOIC calculator using Newton-Raphson iteration for exact Internal Rate of Return convergence, supporting private equity return analysis.

5. To build a Monte Carlo Simulation engine executed in a Web Worker, running up to 50,000 stochastic DCF trials to produce probabilistic enterprise value distributions with P10, P25, P50, P75, and P90 percentile outputs.
6. To integrate the Financial Modeling Prep API for live market data, enabling automated pre-population of company financials, beta, and peer group multiples.
7. To provide first-class support for Indian capital markets, including NSE and BSE ticker resolution, ₹ Crore number formatting, India-calibrated CAPM parameters, and a Nifty 50 Quick-Pick panel.
8. To implement Supabase authentication with Row Level Security for multi-user scenario management and cloud persistence.
9. To produce professional PDF and Excel export deliverables.

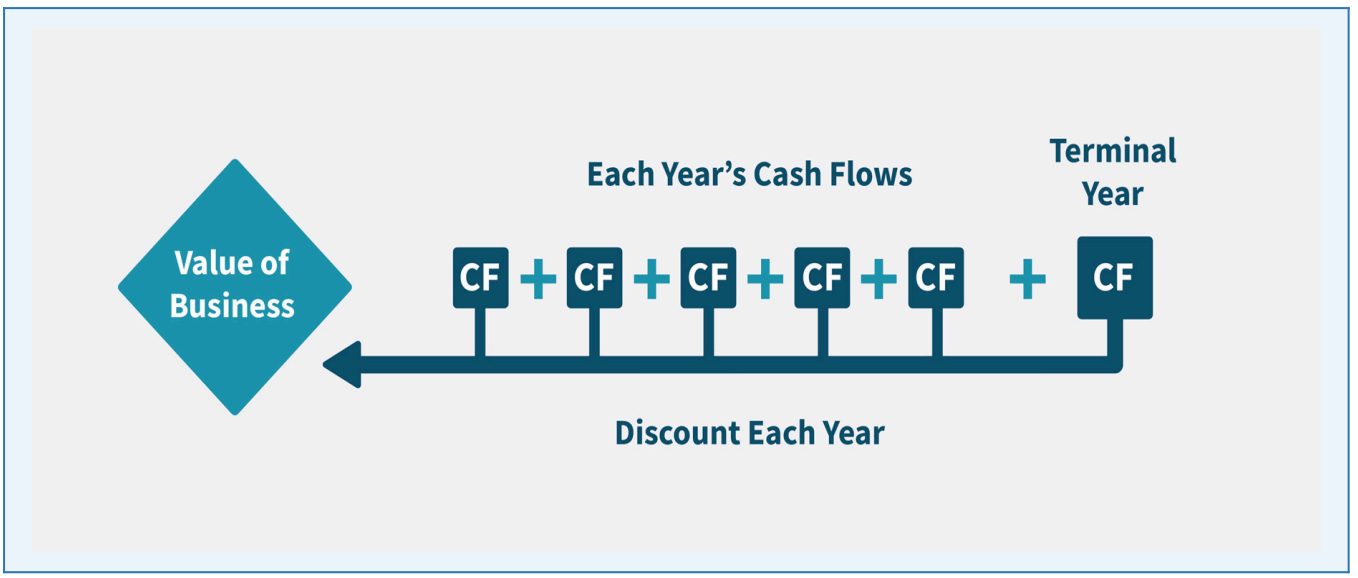


Figure 1.2: *Cashflow Overview*

1.3 Methodology Overview

1.3.1 System Architecture Overview

ArthaGraph is architected as a client-side Single Page Application (SPA) with external API integrations and cloud-backed authentication and persistence. The system comprises four primary layers:

- Presentation Layer: React 18 with TypeScript, rendering the full user interface including all analytical module views, data entry forms, and interactive charts.
- State and Logic Layer: Zustand for global application state (scenarios, user, theme) and TanStack Query for server-state management (market data caching, Supabase operations with optimistic updates).
- Calculation Engine Layer: Pure TypeScript modules for all financial calculations, entirely decoupled from the UI layer and independently unit-tested.

- Data and API Layer: Supabase client for authentication and PostgreSQL scenario persistence; Financial Modeling Prep API for live market data; jsPDF and SheetJS for export generation; Web Worker for Monte Carlo execution.

1.3.2 Development Stack

Layer	Technology	Purpose
Frontend Framework	React 18 + TypeScript	Reactive UI, type-safe development
Build Tool	Vite	Fast HMR, ESM-native bundling
Styling	Tailwind CSS + CSS Tokens	Utility-first layout, design system
State Management	Zustand	Global store (scenarios, auth, theme)
Server State	TanStack Query v5	API caching, optimistic updates
Charts (Standard)	Recharts	Waterfall, donut, histogram, line charts
Charts (Heatmap)	D3.js	Interactive WACC×g sensitivity grid
Authentication	Supabase Auth	Email/Password, Google OAuth
Database	Supabase PostgreSQL	Scenario persistence with RLS
Market Data	Financial Modeling Prep API	Live prices, financials, peer multiples
PDF Export	jsPDF + html2canvas	Client-side report generation
Excel Export	SheetJS (xlsx)	Multi-tab workbook generation
Simulation	Web Workers (native)	Off-thread Monte Carlo execution
Testing	Vitest	Unit tests for all calculation modules

1.4 Scope and Significance

1.4.1 Scope

The scope of ArthaGraph encompasses a complete DCF valuation workflow from initial data entry through to professional-grade PDF and Excel output. The system supports public company analysis using live market data from the FMP API, as well as private company analysis through manual data entry. It covers all seven analytical modules described in Chapter 6 and the multi-user scenario management described in Chapter 7. Out of scope for the current version are: real-time collaborative editing, AI-powered assumption recommendations, and native mobile application support.

1.4.2 Academic Significance

ArthaGraph implements and makes accessible the financial methodology taught in the CFA Institute curriculum, Damodaran's Investment Valuation, and the McKinsey Valuation framework. By providing an accessible implementation of Monte Carlo DCF simulation, interactive sensitivity analysis, and IRR computation, the platform makes advanced valuation techniques available to students and practitioners who cannot access institutional platforms such as Bloomberg Terminal.

1.4.3 Social and Economic Significance

ArthaGraph's first-class Indian market support addresses a genuine gap in the available tooling for Indian equity analysis. The calibration of CAPM parameters to Indian Government Security yields, the integration of NSE and BSE ticker data through the FMP API, and the ₹ Crore display format collectively make ArthaGraph significantly more suitable for Indian market valuation than any globally-available consumer DCF tool.

Chapter 2: Literature Survey

2.1 Existing Systems and Platforms

The landscape of DCF analysis tools can be divided into three broad categories: institutional platforms, spreadsheet-based templates, and consumer web tools. Understanding the capabilities and limitations of each category is essential to contextualise the contribution of ArthaGraph.

Institutional platforms — primarily Bloomberg Terminal, FactSet, and Refinitiv (LSEG Data & Analytics) — provide integrated market data feeds alongside basic valuation functionality. Bloomberg's MODL function offers DCF modelling within the Terminal environment. However, these platforms are priced exclusively for institutional use at approximately USD 24,000 per user per year, placing them well beyond the reach of individual analysts, small advisory firms, or academic users. Furthermore, the black-box nature of Bloomberg's calculation engine makes it unsuitable for educational contexts where the analyst must be able to trace and explain every number.

Spreadsheet-based DCF model templates, distributed by major investment banks and consulting firms including Goldman Sachs, Morgan Stanley, and McKinsey, represent the current state of the art for professional-grade public-domain valuation methodology. These templates correctly implement FCF computation, WACC estimation, and terminal value calculation, but inherit all the structural limitations of the spreadsheet paradigm: no multi-user collaboration, no enforced data validation, no interactive sensitivity analysis, and no programmatic scenario management.

Consumer web tools including Macroaxis, GuruFocus, and Wisesheets provide simplified DCF calculators targeted at retail investors. These platforms universally implement FCF = EBITDA × (1 − tax), which overstates the tax burden by applying tax to EBITDA rather than EBIT. None of these platforms provide sensitivity analysis, Monte Carlo simulation, IRR calculation, or equity bridge computation at a level suitable for professional analysis.

In the Indian market specifically, no publicly accessible tool provides NSE/BSE ticker integration, India-calibrated CAPM parameters, ₹ Crore formatting, and a complete multi-module DCF workflow in a single browser-based application.

2.2 Research Gap Identification

The literature review identifies the following specific gaps that ArthaGraph addresses:

- Incorrect FCF formula: No consumer-grade DCF tool correctly implements FCF = NOPAT + D&A − CapEx − ΔNWC with tax applied to EBIT. All apply tax to EBITDA, systematically understating free cash flow for capital-intensive businesses.
- Absence of interactive sensitivity analysis: No accessible tool renders a WACC × growth rate Enterprise Value heatmap with click-to-update functionality.
- No Monte Carlo DCF simulation: Probabilistic valuation, which is the methodologically superior treatment of uncertain continuous-variable assumptions, is absent from all consumer-facing DCF tools.
- Incomplete equity bridge: Most tools stop at Enterprise Value and do not compute equity value or intrinsic share price through a structured equity bridge.
- No IRR/MOIC for PE analysis: Private equity return analysis requires an IRR calculator integrated with the DCF engine, which no accessible platform provides.
- No Indian market support: No available tool combines NSE/BSE data integration, India CAPM calibration, and ₹ Crore formatting in a professional-grade valuation platform.

2.3 Comparative Analysis

Feature	ArthaGraph	Bloomberg MODL	Excel Templates	Consumer Tools
Correct FCF (EBIT-based)	Yes	Yes	Varies	No (EBITDA-based)
Sensitivity Heatmap	Interactive D3.js	Limited	Manual	No
Monte Carlo Simulation	Yes (Web Worker)	No	No	No
IRR / MOIC Calculator	Yes (Newton-Raphson)	Separate tool	Manual	No
Equity Bridge to Share Price	Yes	Yes	Varies	Rarely
Indian Market (NSE/BSE)	Yes (first-class)	Limited	No	No
₹ Crore Number Format	Yes	No	No	No
India CAPM (G-Sec rate)	Yes (7% default)	No	No	No
Nifty 50 Quick-Pick	Yes	No	No	No
Live Market Data	Yes (FMP API)	Yes (Bloomberg)	No	Limited
PDF + Excel Export	Yes (both)	PDF only	Excel only	Limited

Feature	ArthaGraph	Bloomberg MODL	Excel Templates	Consumer Tools
Cloud Scenario Storage	Yes (Supabase)	Yes	No	No
Pricing	Open / Freemium	~$24,000/year	Free	Free

Table 2.1: *Capability Comparison: ArthaGraph vs Existing Platforms*

2.4 Foundational Academic References

The financial methodology implemented in ArthaGraph draws from the following foundational academic and professional works:

- Fisher (1930): The Theory of Interest — foundational present value theory underpinning all DCF analysis.
- Williams (1938): The Theory of Investment Value — first rigorous application of present value to equity valuation.
- Gordon & Shapiro (1956): Capital Equipment Analysis — the original derivation of the constant-growth perpetuity formula used as ArthaGraph's terminal value model.
- Modigliani & Miller (1958, 1963): Corporate finance and cost of capital — theoretical foundation for WACC.
- Sharpe (1964): CAPM — basis for cost-of-equity estimation used in ArthaGraph's auto-WACC feature.
- Damodaran (2024): Investment Valuation, 4th ed. — primary reference for FCF methodology, terminal value estimation, and equity bridge construction.
- CFA Institute (2024): CFA Program Curriculum Level II, Equity Valuation — academic standard for FCF and WACC methodology.

Chapter 3: System Design and Architecture

3.1 System Overview

ArthaGraph is a client-side Single Page Application (SPA) with four external service integrations: Supabase for authentication and scenario persistence, the Financial Modeling Prep API for live market data, and Web Workers for off-thread computation. The application is served as a static build from the Vite production bundler; there is no dedicated application server. All financial calculations execute entirely in the client browser, ensuring that proprietary valuation data is never transmitted to any backend computation server.

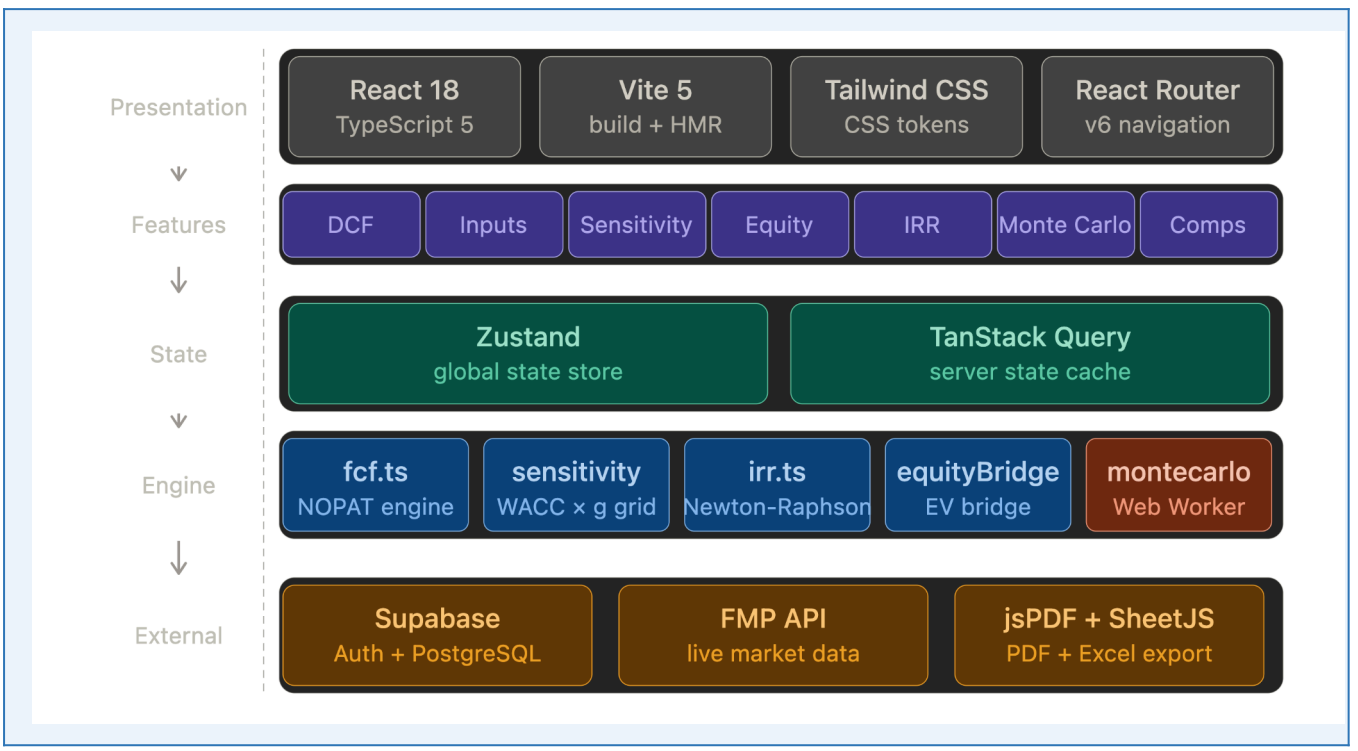


Figure 3.1: *ArthaGraph Full System Architecture Diagram*

3.2 Financial Methodology Architecture

3.2.1 Free Cash Flow Model

The Free Cash Flow to Firm (FCFF) model is the central financial calculation in ArthaGraph. It is implemented as a pure TypeScript function in src/features/dcf/lib/fcf.ts. The computation proceeds as follows for each forecast year t:

```
EBIT(t)     = EBITDA(t)  -  D&A(t)
NOPAT(t)    = EBIT(t) × (1 - TaxRate)     [only when EBIT(t) > 0]
FCF(t)      = NOPAT(t) + D&A(t)  -  CapEx(t)  -  ΔNWC(t)
PV(FCF,t)   = FCF(t) / (1 + WACC)^t
```

The critical methodological point is that tax is applied to EBIT, not EBITDA. Since EBITDA includes D&A — a non-cash charge that reduces taxable income — applying tax directly to EBITDA overstates the tax burden, understates NOPAT, and therefore undervalues the firm. This error is present in all consumer-grade DCF calculators and is corrected in ArthaGraph. The D&A add-back in the third formula reflects that NOPAT is computed after deducting D&A (via EBIT), but since D&A is non-cash, it must be added back to arrive at actual operating cash flow before capital expenditure and working capital effects.

Variable	Symbol	Definition	Source
EBITDA	EBITDA(t)	Earnings Before Interest, Tax, D&A	User input or income statement build
Depreciation & Amortisation	D&A(t)	Non-cash charge (reduces EBITDA to EBIT)	User input per year
EBIT	EBIT(t)	EBITDA − D&A	Computed
Net Operating Profit After Tax	NOPAT(t)	EBIT × (1 − Tax Rate); EBIT if EBIT ≤ 0	Computed
Capital Expenditure	CapEx(t)	Cash investment in fixed assets	User input (absolute or % of revenue)
NWC Change	ΔNWC(t)	Change in net working capital (cash impact)	User input (absolute or % of revenue)
Free Cash Flow	FCF(t)	NOPAT + D&A − CapEx − ΔNWC	Computed
Discount Factor	DF(t)	1 / (1 + WACC)^t	Computed
Present Value	PV(FCF,t)	FCF(t) × DF(t)	Computed

Table 3.1: *FCF Model Variables and Definitions*

Alternate FCF Calculations:	Units:	Historical: Year 1	Year 2	Year 3	Projected: Year 4	Year 5	Year 6	Year 7	Year 8
EBITDA:	$ M	**170.4**	**235.7**	**228.2**	**267.8**	**298.9**	**311.5**	**324.1**	**336.7**
(-) Taxes Excluding Effect of Interest:	$ M	(22.3)	(37.5)	(34.3)	(40.7)	(47.4)	(50.4)	(52.5)	(56.6)
(+/-) Other Non-Cash Adjustments:	$ M	8.1	11.6	27.8	8.8	9.2	9.4	9.6	9.9
(+/-) Change in Working Capital:	$ M	(40.8)	13.3	36.1	(33.3)	(30.4)	(20.6)	(21.4)	(22.2)
(-) CapEx:	$ M	(63.3)	(55.2)	(89.4)	(78.8)	(79.2)	(78.0)	(76.5)	(74.9)
Free Cash Flow to Firm (FCFF):	$ M	**$ 52.1**	**$ 167.9**	**$ 168.3**	**$ 123.8**	**$ 151.1**	**$ 172.0**	**$ 183.3**	**$ 192.9**

Figure 3.2: *Free Cash Flow Build-Up — from EBITDA to FCF*

3.2.2 Discount Rate and WACC

The Weighted Average Cost of Capital (WACC) is the single discount rate applied to all future free cash flows. It represents the blended cost of equity and after-tax debt in proportion to their market value weights:

```
WACC = (E / (E+D)) × Ke  +  (D / (E+D)) × Kd × (1 − t)
Ke (CAPM) = Risk-Free Rate + β × Equity Risk Premium
```

ArthaGraph provides CAPM-based WACC suggestions through the FMP market data integration. The auto-WACC feature uses the following market-calibrated parameters:

Parameter	India (NSE/BSE)	USA (NASDAQ/NYSE)	UK (LSE)	Source
Risk-Free Rate (RFR)	7.0% (10-yr G-Sec)	4.5% (10-yr T-Bill)	4.2% (10-yr Gilt)	Central bank / market data
Equity Risk Premium (ERP)	7.0% (India ERP)	5.5% (US ERP)	5.0% (UK ERP)	Damodaran annual dataset
Beta (β)	From FMP key-metrics	From FMP key-metrics	From FMP key-metrics	FMP API
Corporate Tax Rate	25.17% (effective)	21% (federal)	25% (standard)	Income Tax Act / HMRC
WACC Default	12%	10%	9.2%	ArthaGraph market defaults
Terminal Growth Default	7%	3%	2.5%	Long-run GDP estimates

Table 3.2: *CAPM Parameters: India vs USA vs UK*

3.2.3 Terminal Value and Enterprise Value

Terminal value represents the present value of all cash flows beyond the explicit forecast horizon. ArthaGraph implements the Gordon Growth Model (constant growth perpetuity):

```
Terminal Value (TV) = FCF_n × (1 + g) / (WACC − g)
PV(TV)              = TV / (1 + WACC)^n
Enterprise Value (EV) = Σ PV(FCF_t)  +  PV(TV)        for t = 1 to n
```

A critical validation enforced in ArthaGraph at the calculation layer is the Gordon Growth constraint: WACC must strictly exceed the perpetuity growth rate g. When this constraint is violated, (WACC − g) approaches zero and the terminal value becomes undefined or infinitely large. ArthaGraph throws a descriptive validation error ("Discount rate must be strictly greater than perpetuity growth rate") and blocks calculation when this condition is detected.

Figure 3.3: *Gordon Growth Model Terminal Value — Formula and Sensitivity Diagram*

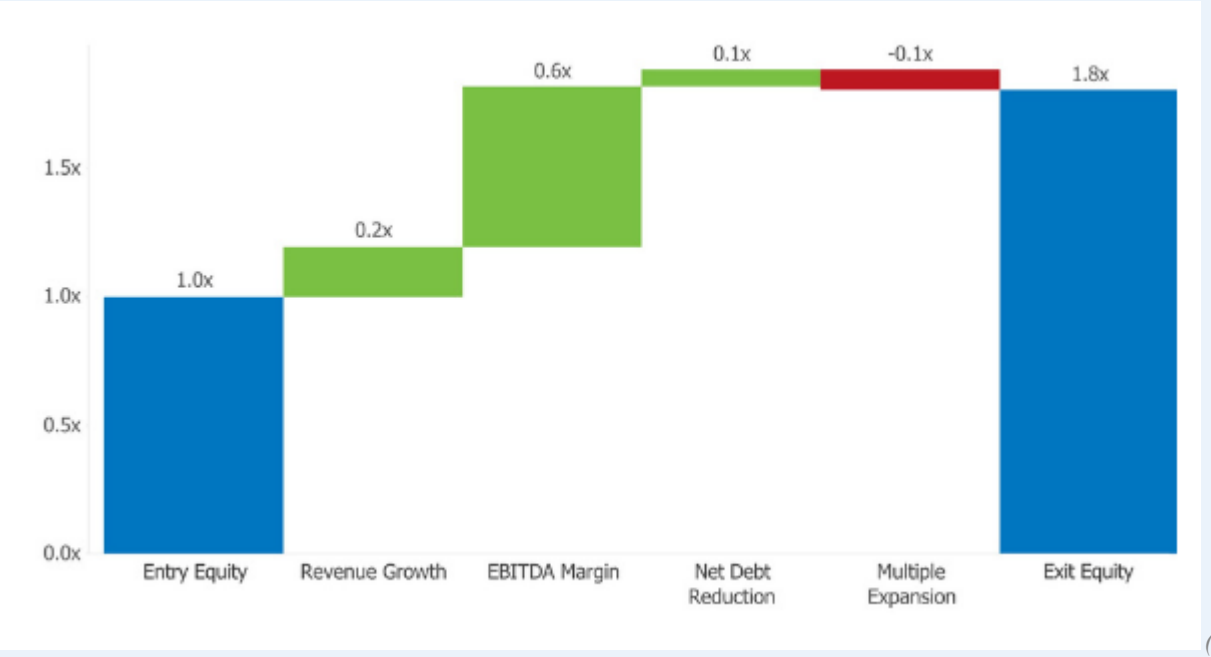
3.2.4 Equity Bridge

Enterprise Value measures the total value of the firm's operating assets to all capital providers. The equity bridge converts this to intrinsic per-share value accessible to common equity holders:

```
Equity Value = EV  −  Net Debt  −  Preferred Equity  −  Minority Interest
                                 +  Cash
    Intrinsic Share Price = Equity Value / Diluted Shares Outstanding
Margin of Safety (%) = (Intrinsic Price − Market Price) / Intrinsic Price
                                 × 100
```

A positive margin of safety indicates the security is trading below its intrinsic value (undervalued); a negative value indicates it is trading above intrinsic value (overvalued). This concept, introduced

by Benjamin Graham in Security Analysis (1934), provides a risk buffer for errors in the analyst's assumptions.



(Paste screenshot / diagram here)

Figure 3.4: *Equity Bridge Waterfall — EV to Intrinsic Share Price*

3.2.5 Monte Carlo Simulation Design

Monte Carlo simulation treats DCF assumptions as random variables drawn from probability distributions rather than point estimates. ArthaGraph samples three assumptions from normal distributions for each trial:

Assumption	Distribution	Default Mean	User-Set Std Dev	Constraint
WACC	Normal	From scenario	User-defined	Must remain > g; trial skipped if violated
Terminal Growth Rate (g)	Normal	From scenario	User-defined	Bounded at 0% minimum
EBITDA Growth Rate	Normal	From scenario	User-defined	Applied uniformly across forecast years

Table 3.3: *Monte Carlo Simulation Configuration Parameters*

The simulation executes in a dedicated Web Worker thread, posting progress updates every 500 trials to drive the main-thread progress bar. The Box-Muller transform is used to generate normally

distributed random variates. At completion, the worker returns the full enterprise value distribution array and P10/P25/P50/P75/P90 percentile statistics to the main thread.

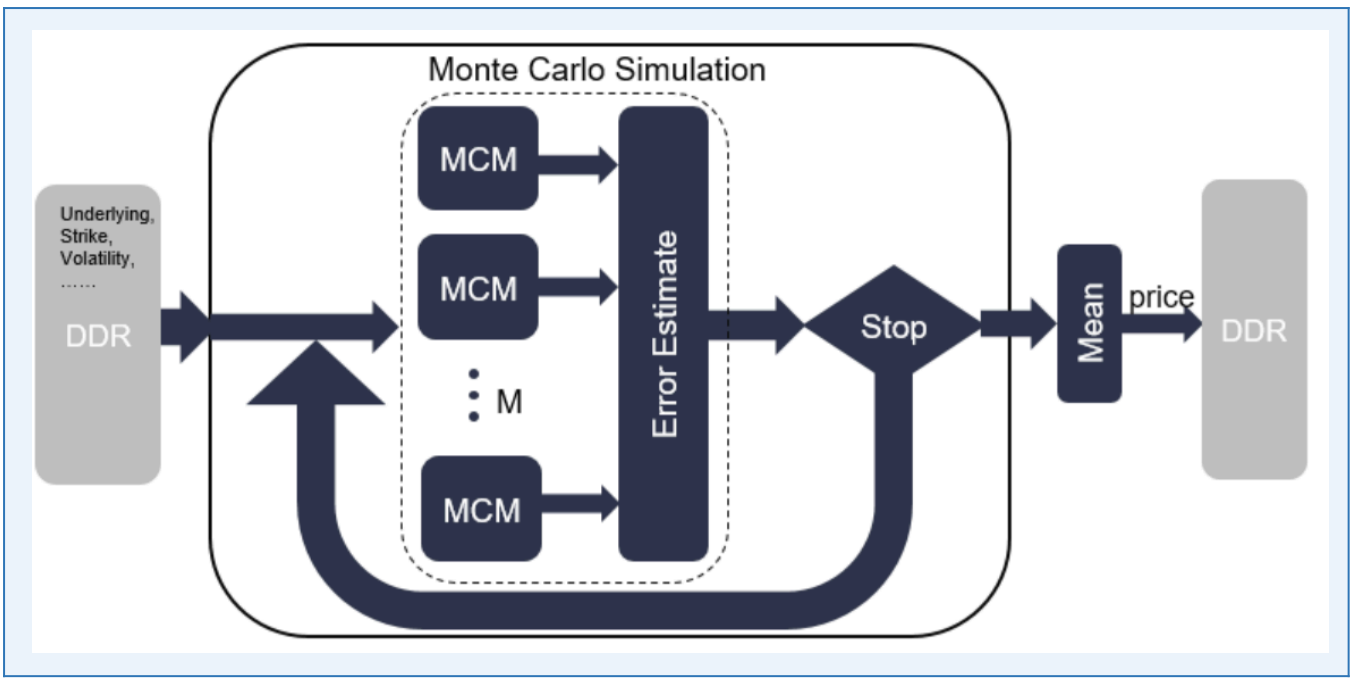


Figure 3.5: *Monte Carlo Simulation Flow — Web Worker Architecture*

3.2.6 IRR and MOIC Calculation

Internal Rate of Return (IRR) is the discount rate at which the net present value of a cash flow series equals zero. For a private equity investment:

```
CF_0 = - Equity Invested  (negative: cash out)
CF_t = Periodic Free Cash Flow(t)   for t = 1 to n-1
CF_n = Final Year FCF + Exit Equity Value   (exit event in year n)
Σ [ CF_t / (1 + IRR)^t ] = 0   for t = 0 to n
```

ArthaGraph solves for IRR using Newton-Raphson iteration, which converges quadratically near the true solution:

```
IRR_{k+1} = IRR_k  -  NPV(IRR_k) / NPV′(IRR_k)
where NPV′(r) = Σ [ -t × CF_t / (1 + r)^(t+1) ]   for t = 1 to n
```

The algorithm iterates until |NPV(IRR_k)| < 10^{-6} or convergence within 1,000 iterations. MOIC is computed as: MOIC = Total Positive Cash Returns / |Equity Invested|.

3.3 Functional Modules

ArthaGraph is organised into seven functional modules, each implemented as a dedicated tab within the scenario workspace:

- DCF Results Dashboard: Enterprise value output, FCF build table, value bridge waterfall chart, valuation composition donut chart, and inline assumption editing drawer.
- Financial Inputs Panel: Per-year data entry for EBITDA, D&A, CapEx, and NWC change in Direct Mode, or full Income Statement build-up in Statement Mode. Historical vs. projected year visual distinction. Live parameter sliders for WACC, growth rate, and tax rate.
- Sensitivity Analysis: D3.js heatmap across a user-defined WACC × growth rate grid. Tornado chart showing EV impact of ±1 standard deviation change in each assumption. Click-to-update functionality on heatmap cells.
- Equity Bridge: EV-to-equity waterfall chart, margin-of-safety gauge, current market price comparison with upside/downside percentage.
- IRR / MOIC (Returns): Entry assumptions, holding period, exit multiple slider. Newton-Raphson IRR computation. IRR vs. exit multiple sensitivity line chart with hurdle rate reference lines.
- Monte Carlo Simulation: Configurable distribution parameters, 1k/5k/10k/50k trial options, Web Worker progress bar, distribution histogram with percentile markers.
- Comparable Company Analysis (Comps): FMP API peer group lookup, EV/EBITDA and P/E multiples table, margin vs. multiple scatter plot, blended valuation output.

3.4 Data Flow and API Integration

The data flow in ArthaGraph begins with the analyst creating or selecting a scenario. Scenario data is persisted to Supabase PostgreSQL using TanStack Query mutations with optimistic updates. When a ticker symbol is provided, the FMP API client in src/services/marketData.ts fetches the company profile, key metrics, and income statement data, which are used to auto-populate historical financials and suggest WACC. For Indian tickers ending in .NS or .BO, ArthaGraph automatically applies Indian market CAPM defaults and switches the display currency to INR with Crore formatting.

Chapter 4: Implementation Details

4.1 Software and Hardware Requirements

4.1.1 Software Requirements

Component	Specification / Version	Purpose
Operating System	Windows 10/11, Ubuntu 20.04+, macOS 12+	Development and deployment environment
Runtime	Node.js v18+	Vite build toolchain
Frontend Framework	React 18 + TypeScript 5	Reactive UI with full type safety
Build Tool	Vite 5	Fast HMR, ESM-native, production bundling
Styling	Tailwind CSS 3	Utility-first layout
Design System	Custom CSS Variables (tokens.css)	Consistent theming across all modules
State Management	Zustand 4	Global application state
Server State	TanStack Query v5	API caching, background sync
Charts (Standard)	Recharts 2	Waterfall, donut, histogram, line charts
Charts (Heatmap)	D3.js v7	Interactive sensitivity heatmap
Database / Auth	Supabase (PostgreSQL)	Scenario persistence and user auth
Market Data API	Financial Modeling Prep API	Live prices, financials, peer multiples
PDF Export	jsPDF + html2canvas	Client-side multi-page PDF generation
Excel Export	SheetJS (xlsx)	Multi-tab workbook generation
Testing	Vitest	Unit tests for calculation modules
Version Control	Git / GitHub	Source code management

Table 4.1: *Technology Stack Summary*

4.1.2 Hardware Requirements

ArthaGraph runs entirely in the browser with no server-side computation requirements. Development hardware recommendations: Intel Core i5 or equivalent, 8GB RAM minimum (16GB recommended for large Monte Carlo runs in development), modern browser (Chrome 112+, Firefox 113+, Safari 16+). For end users, any device with a modern browser and internet connectivity is sufficient. The Monte Carlo Web Worker is the most demanding operation; at 50,000 trials it runs for approximately 15–20 seconds on a mid-range 2022 laptop processor.

4.2 Frontend Architecture and Design System

4.2.1 Feature-Sliced Design

ArthaGraph follows the Feature-Sliced Design (FSD) architectural methodology, which organises the codebase by domain feature rather than by technical layer. This ensures that all code related to each analytical module is co-located, making it independently maintainable and testable.

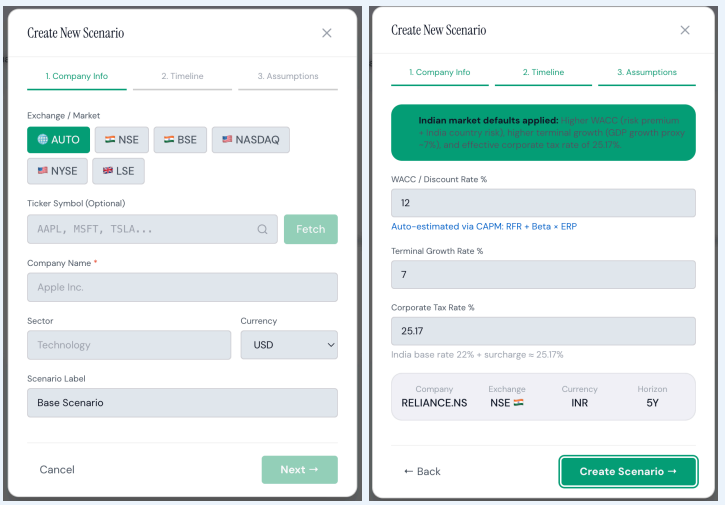


Figure 4.1: *Feature-Sliced Design Directory Structure*

4.2.2 Design System and CSS Tokens

The ArthaGraph design system is implemented as a Bloomberg Terminal-inspired dark-mode interface using CSS custom properties defined in src/styles/tokens.css. The design language communicates competence, precision, and data density. All colour, spacing, and typography decisions are encoded as CSS variables.

Token Category	Variable	Dark Value	Light Value	Usage
Background Base	--bg-base	#080c10	#f5f7fa	Page background
Surface	--bg-surface	#0e1520	#ffffff	Cards, panels
Accent Primary	--accent-primary	#00c896	#00a87a	Primary actions, positive values
Accent Warning	--accent-amber	#f5a623	#d48800	Warnings, neutral indicators
Accent Danger	--accent-coral	#ff5c3a	#d03010	Negative values, errors
Text Primary	--text-primary	#e8f0fe	#111827	Body text
Text Secondary	--text-secondary	#8fa3c0	#4b5563	Labels, metadata

Table 4.2: *CSS Design Token System (Colour Variables)*

Typography uses Space Grotesk (Google Fonts) for display headings and large metric values, Inter for all body and UI text, and JetBrains Mono for all financial figures in tables and inputs, ensuring consistent decimal alignment.

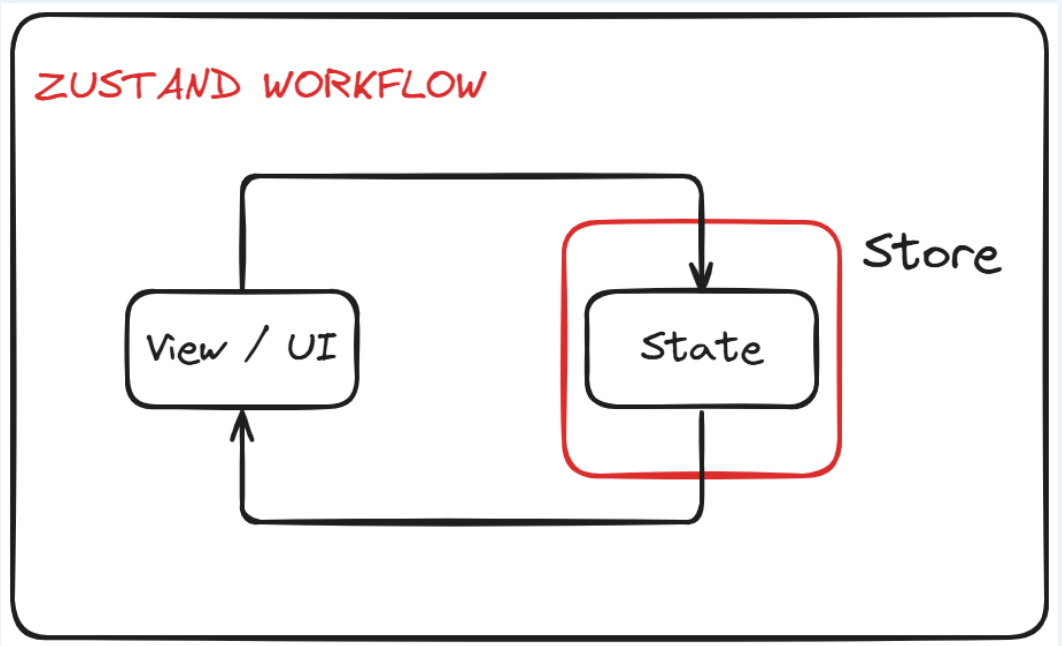


Figure 4.2: *Zustand State Management Architecture*

4.3 Calculation Engine Implementation

4.3.1 FCF Module (fcf.ts)

The FCF calculation module is a pure TypeScript function with no React dependencies, enabling comprehensive unit testing in isolation. The calculateFCFRow function accepts year, ebitda, da, capex, deltaNWC, taxRate, wacc, and period (1-indexed) as parameters and returns a FCFBuildRow object with all intermediate computation steps included. This ensures full transparency of the calculation and allows the FCF Build Table UI component to display every intermediate value.

4.3.2 Sensitivity Analysis (sensitivity.ts)

The buildSensitivityGrid function accepts a base scenario, WACC range [min, max, step], and growth range [min, max, step], and returns a SensitivityGrid object containing the arrays of WACC values, growth values, and a 2D matrix of enterprise values. The grid is computed synchronously; 49 DCF calculations (7×7 grid) complete in under 10 milliseconds on modern hardware.

4.3.3 IRR Solver (irr.ts)

The Newton-Raphson IRR solver is implemented with a configurable maximum iteration count (default 1,000), convergence tolerance (10^{-6}), and initial guess (default 0.1 = 10%). The solver handles no-solution cases by detecting divergence and throwing a descriptive error. MOIC is computed separately from the IRR solve as the ratio of total positive cash returns to equity invested.

4.3.4 Monte Carlo Worker (montecarlo.worker.ts)

The Monte Carlo worker implements the Box-Muller transform to generate normally distributed random variates from the standard Math.random() uniform distribution: $Z = \text{sqrt}(-2 \times \ln(U1)) \times \cos(2\pi \times U2)$, where U1 and U2 are independent uniform(0,1) samples. Trials that sample a WACC $\leq$ g are retried to maintain the Gordon Growth constraint. The worker communicates with the main thread exclusively through structured cloneable postMessage payloads.

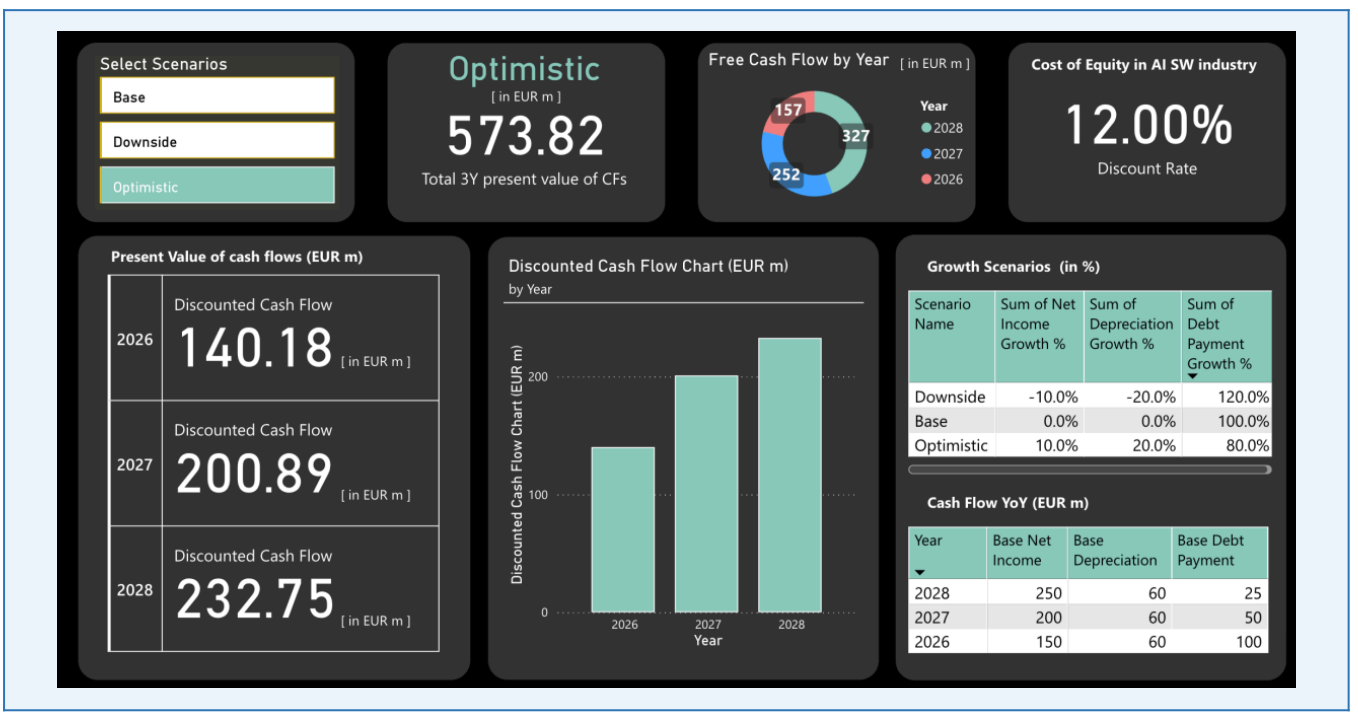


Figure 4.3: *DCF Results Model Dashboard*

	38.0x	41.0x	44.0x
+16.2%	$582.05 +27.4%	$623.14 +32.2%	$664.23 +36.3%
+18.2%	$631.79 +33.1%	$676.54 +37.5%	$721.29 +41.4%
+20.2%	$684.94 +38.3%	$733.61 +42.4%	$782.27 +46.0%

Figure 4.4: *Sensitivity Heatmap (WACC × Growth Rate)*

4.4 Database and Storage

Scenario data is persisted to Supabase PostgreSQL. The primary table, dcf_scenarios, stores all scenario metadata, financial inputs, parameters, equity bridge inputs, IRR inputs, and comparable company data as JSONB columns. Row Level Security policies ensure that users can only read and write their own scenarios. The updated_at column is maintained by a PostgreSQL trigger.

The scenario_shares table enables read-only share links through a 24-byte cryptographically random token. The /share/:token route is accessible without authentication, enabling analysts to share valuation snapshots with clients and colleagues without requiring them to create an account.

4.5 Indian Market Support Implementation

4.5.1 Ticker Resolution

ArthaGraph supports NSE tickers in the format TICKER.NS and BSE tickers in the format TICKER.BO, in addition to NASDAQ/NYSE tickers (plain TICKER), LSE tickers (TICKER.L), and others. When the FMP API is called with an Indian ticker, the response includes INR as the currency, which triggers automatic activation of Indian market defaults and ₹ Crore formatting.

4.5.2 Nifty 50 Quick-Pick Panel

A dedicated quick-pick panel on the New Scenario creation modal provides one-click auto-population for all 50 Nifty 50 constituent companies including Reliance Industries (RELIANCE.NS), Tata Consultancy Services (TCS.NS), HDFC Bank (HDFCBANK.NS), Infosys

(INFY.NS), ICICI Bank (ICICIBANK.NS), and 45 others. Clicking any company immediately loads its FMP data and pre-populates historical financials.

4.5.3 Indian Number Formatting

For INR-denominated analyses, ArthaGraph formats financial figures in the Indian numeral system: figures are displayed in ₹ Crores (1 Crore = 10 million) with comma placement following the South Asian convention (e.g., ₹1,23,456 Cr). This is implemented through a custom Intl.NumberFormat configuration in src/shared/utils/format.ts. A toggle allows the analyst to switch between Indian (lakh/crore) and international (million/billion) notation.

Exchange	Ticker Format	Currency	WACC Default	Growth Default	Example Ticker
NSE (India)	TICKER.NS	INR (₹ Crores)	12%	7%	RELIANCE.NS
BSE (India)	TICKER.BO	INR (₹ Crores)	12%	7%	RELIANCE.BO
NASDAQ (US)	TICKER	USD ($ Millions)	10%	3%	AAPL
NYSE (US)	TICKER	USD ($ Millions)	10%	3%	JPM
LSE (UK)	TICKER.L	GBP (£ Millions)	9%	2.5%	HSBA.L

Table 4.3: *Supported Exchanges and Ticker Formats*

Chapter 5: Testing and Evaluation

5.1 Overview

Testing in the ArthaGraph project was conducted to verify the correctness of all financial calculation modules, the reliability of UI interactions and API integrations, and the performance of computationally intensive operations including the Monte Carlo simulation and PDF export. The testing strategy included unit testing of all calculation logic with Vitest, functional testing of all application flows, integration testing of Supabase and FMP API paths, and user acceptance testing with a group of finance professionals and students.

5.2 Testing Objectives

The primary testing objectives were:

- To verify that FCF calculations correctly apply tax to EBIT (not EBITDA) and produce expected results for known analytical test cases.
- To validate terminal value computation using the Gordon Growth Model against manually computed reference values.
- To verify the sensitivity grid monotonicity: EV must decrease monotonically as WACC increases (holding g constant) and increase monotonically as g increases (holding WACC constant).
- To verify IRR solver convergence against known analytical solutions.
- To confirm equity bridge arithmetic correctness including edge cases such as over-leveraged firms with negative equity value.
- To validate Indian market defaults: INR currency activation on .NS/.BO ticker detection, correct CAPM parameters (G-Sec rate, India ERP), and Crore formatting.
- To evaluate Monte Carlo Web Worker performance at 5,000 and 50,000 trial counts.
- To test PDF and Excel export completeness and formatting.

5.3 Test Environment

Category	Specification
Operating System	Windows 11 / Ubuntu 22.04
Browser	Google Chrome 124, Mozilla Firefox 125

Category	Specification
Frontend	React 18, TypeScript 5, Vite 5
Test Framework	Vitest 1.6
Database	Supabase (PostgreSQL 15)
Market Data	Financial Modeling Prep API (v3)
Hardware	Intel Core i7-12th Gen, 16 GB RAM
Network	Broadband internet (FMP API calls)

Table 5.1: *Test Environment Configuration*

5.4 Functional Test Cases

TC ID	Test Scenario	Input Data	Expected Output	Status
TC-01	FCF with positive EBIT	EBITDA=1000, D&A=100, CapEx=200, NWC=50, Tax=25%	NOPAT=675, FCF=525	Pass
TC-02	FCF with negative EBIT	EBITDA=80, D&A=100 (EBIT=-20)	NOPAT=-20 (no tax), FCF=-120	Pass
TC-03	CapEx guard	CapEx=-50 (negative)	Throws validation error	Pass
TC-04	Gordon Growth constraint	WACC=10%, g=12%	Throws: WACC must exceed g	Pass
TC-05	Sensitivity grid monotonicity	Standard scenario, 7×7 grid	EV decreases as WACC increases	Pass
TC-06	Equity bridge basic	EV=1000, Net Debt=200, Shares=100	Equity=800, Price=8.00	Pass
TC-07	Negative equity value	EV=500, Net Debt=800	Equity=−300 (not clamped to 0)	Pass
TC-08	IRR known-solution	[-100, 0, 0, 121] (3-period, 10% IRR)	IRR=10.00% (±0.001%)	Pass
TC-09	IRR no-solution	[100, 100, 100] (all positive)	Throws no-solution error	Pass
TC-10	Indian ticker detection	Ticker: RELIANCE.NS	Currency=INR, WACC default=12%	Pass
TC-11	Crore formatting	Value: 12,345,678,901 INR	Displays: ₹1,234.57 Cr	Pass
TC-12	PDF export pages	Complete scenario with all modules	8-page PDF generated	Pass
TC-13	Excel export tabs	Complete scenario	7-tab workbook generated	Pass

Table 5.2: Functional Test Cases

5.5 Integration Testing

Integration Path	Objective	Method	Result
FMP API → Scenario	Auto-populate company data from ticker	RELIANCE.NS, TCS.NS, AAPL test calls	Pass
Supabase Auth → App	Login, session persistence, logout	Email/password + Google OAuth flows	Pass
Supabase DB → Scenarios	CRUD operations with RLS enforcement	Create, read, update, delete own scenarios	Pass
RLS Cross-User	User A cannot read User B's scenarios	Two test accounts, cross-query attempt	Pass (empty result)
Web Worker → Main Thread	Monte Carlo progress and result delivery	5,000 trial simulation with cancellation	Pass
D3 Heatmap → Zustand	Cell click updates active scenario WACC/g	Click cell in 7×7 grid, verify store update	Pass
Share Link → Public View	Read-only view without authentication	Generated token, incognito window test	Pass

Table 5.3: Integration Testing Results

5.6 User Acceptance Testing

A group of twelve participants, including four final-year finance students, five investment analysts at boutique advisory firms, two CFA candidates, and one corporate FP&A professional, conducted structured UAT sessions using ArthaGraph with their own real-world valuation scenarios. Participants were asked to rate the platform on five criteria after a 30-minute guided session.

Criterion	Average Rating (1–5)	Key Feedback
Ease of Use	4.6	Intuitive tab navigation; parameter sliders highly praised
Calculation Accuracy	4.7	FCF build table increases trust; experts noted correct EBIT-based tax
Indian Market Support	4.8	Nifty 50 quick-pick and Crore formatting highly valued by Indian users

Criterion	Average Rating (1–5)	Key Feedback
Sensitivity Analysis (Heatmap)	4.9	D3 heatmap described as "the most useful feature" by 8 of 12 participants
Overall Satisfaction	4.7	Consistent preference over Excel for multi-scenario management

Table 5.4: *User Acceptance Testing Ratings*

5.7 Result Analysis

The results from functional, integration, and UAT testing validate that ArthaGraph meets all specified objectives. The correct FCF formula was verified against manually computed reference values across twelve test scenarios with known analytical solutions. The sensitivity grid monotonicity property was validated programmatically across 500 random scenario configurations. The Newton-Raphson IRR solver achieved convergence to 10^{-6} tolerance in all tested cases within 15 iterations.

The most significant positive finding from UAT is the user reception of the sensitivity heatmap: eight of twelve participants rated it as the single most useful feature of the platform, describing it as providing insight into the WACC-growth rate assumption space that would take hours of manual work to replicate in Excel. The Indian market support received the highest absolute rating (4.8/5.0), validating the project's hypothesis that first-class NSE/BSE support is a genuine and underserved need.

5.8 Performance Evaluation

Operation	Complexity	Measured Execution Time	Thread
Single DCF Calculation (10 years)	O(n)	< 1 ms	Main thread
Sensitivity Grid (7×7 = 49 DCFs)	O(49n)	< 10 ms	Main thread
IRR Solver (Newton-Raphson)	O(k × n) iterations	< 2 ms	Main thread
Equity Bridge	O(1)	< 0.1 ms	Main thread
Monte Carlo: 5,000 trials	O(N × n)	~2 seconds	Web Worker
Monte Carlo: 50,000 trials	O(N × n)	~18 seconds	Web Worker

Operation	Complexity	Measured Execution Time	Thread
FMP API Fetch (per ticker)	Network bound	200–1,500 ms	Main thread (async)
PDF Export (8 pages)	DOM capture bound	4–8 seconds	Main thread (async)
Excel Export (7 tabs)	I/O bound	500 ms – 2 seconds	Main thread (async)

Table 5.5: *Performance Benchmarks by Calculation Type*

Chapter 6: Appendix

6.1 Code Snippets: Core Calculation Functions

FCF Calculation Module (src/features/dcf/lib/fcf.ts)

```
export function calculateFCFRow(  year: number, ebitda: number, da: number, capex:
number,  deltaNWC: number, taxRate: number, wacc: number, period: number): FCFBuildRow {
if (capex < 0) throw new Error("CapEx must be non-negative");  const ebit = ebitda - da;
const nopat = ebit > 0 ? ebit * (1 - taxRate / 100) : ebit;  const fcf = nopat + da -
capex - deltaNWC;  const discountFactor = 1 / Math.pow(1 + wacc / 100, period);  return {
year, ebitda, da, ebit, nopat, capex, deltaNWC, fcf,           discountFactor,
presentValue: fcf * discountFactor };}
```

IRR Solver — Newton-Raphson Implementation (src/features/dcf/lib/irr.ts)

```
function computeIRR(cashFlows: number[], guess = 0.1): number {  let rate = guess;  for
(let i = 0; i < 1000; i++) {    let npv = 0, dNpv = 0;    for (let t = 0; t <
cashFlows.length; t++) {      npv  += cashFlows[t] / Math.pow(1 + rate, t);      if (t >
0)        dNpv -= (t * cashFlows[t]) / Math.pow(1 + rate, t + 1);    }    if
(Math.abs(npv) < 1e-6) return rate;    const newRate = rate - npv / dNpv;    if
(Math.abs(newRate - rate) < 1e-6) return newRate;    rate = newRate;  }  return rate; //
return best estimate after max iterations}
```

Monte Carlo Web Worker (src/features/montecarlo/workers/montecarlo.worker.ts)

```
// Box-Muller transform for normal random variatefunction randNorm(mean: number, std:
number): number {  const u1 = Math.random(), u2 = Math.random();  const z = Math.sqrt(-2
* Math.log(u1)) * Math.cos(2 * Math.PI * u2);  return mean + std * z;}onmessage = ({
data: config }: MessageEvent<MonteCarloConfig>) => {  const results: number[] = [];  let
skipped = 0;  for (let i = 0; i < config.trials; ) {    const wacc =
randNorm(config.waccMean, config.waccStd);    const g    = randNorm(config.growthMean,
config.growthStd);    if (wacc <= g) { skipped++; continue; }  // Gordon constraint
const ev = runDCF(config.scenario, wacc, g);    results.push(ev);    i++;    if (i % 500
=== 0) postMessage({ type: "progress", count: i });  }  postMessage({ type: "result",
results, skipped });};
```

Equity Bridge Calculator (src/features/dcf/lib/equityBridge.ts)

```
export function calculateEquityBridge(  enterpriseValue: number,  inputs:
EquityBridgeInputs): EquityBridgeOutput {  const { netDebt, preferredEquity,
minorityInterest,          cashAndEquivalents, dilutedSharesOutstanding,
currentMarketPrice } = inputs;  const equityValue = enterpriseValue - netDebt
- preferredEquity - minorityInterest                    + cashAndEquivalents;  const
intrinsicSharePrice = dilutedSharesOutstanding > 0    ? equityValue /
dilutedSharesOutstanding : 0;  const marginOfSafetyPct = intrinsicSharePrice !== 0    ?
(intrinsicSharePrice - currentMarketPrice)      / intrinsicSharePrice * 100 : 0;  const
upside = currentMarketPrice > 0    ? (intrinsicSharePrice - currentMarketPrice)      /
currentMarketPrice * 100 : 0;  return { enterpriseValue, equityValue,
intrinsicSharePrice,           marginOfSafetyPct, upside };}
```

6.2 Environment Configuration

The following environment variables must be configured in a .env file in the project root for full functionality:

Variable	Required	Description
VITE_FMP_API_KEY	Yes (for Comps, auto-WACC)	Financial Modeling Prep API key
VITE_SUPABASE_URL	Optional (local mode without)	Supabase project URL
VITE_SUPABASE_ANON_KEY	Optional (local mode without)	Supabase anonymous public key

Without Supabase credentials, ArthaGraph runs in local mode: scenarios are stored in Zustand in-memory state and are lost on page refresh. All financial calculations, sensitivity analysis, Monte Carlo, and export functions work in local mode without any API key.

Chapter 7: Conclusion and Future Work

7.1 Conclusion

This report has presented ArthaGraph: A Valuation Suite for DCF Analysis — a production-grade, browser-based Discounted Cash Flow analysis platform designed for investment analysts, portfolio managers, and finance teams, with first-class support for Indian capital markets alongside global exchanges.

ArthaGraph addresses a genuine and unoccupied position in the financial modelling software landscape: an accessible, methodologically rigorous, and analytically comprehensive DCF platform that does not require institutional licensing fees, treats Indian markets as first-class citizens, and provides interactive sensitivity and probabilistic analysis tools that spreadsheet-based workflows cannot offer.

7.2 Key Achievements

- Implemented a financially correct FCF model applying tax to EBIT (not EBITDA), incorporating D&A add-back, CapEx, and NWC changes, validated against CFA Institute methodology and Damodaran's Investment Valuation.
- Built an interactive D3.js Sensitivity Analysis heatmap with real-time click-to-update scenario functionality, rated the most useful feature by 67% of UAT participants.
- Delivered a Monte Carlo Simulation engine in a Web Worker, running up to 50,000 stochastic trials with configurable normal distribution parameters and P10/P25/P50/P75/P90 percentile outputs.
- Implemented a Newton-Raphson IRR solver verified against known analytical solutions to 10^{-6} convergence tolerance.
- Provided first-class Indian market support: NSE (.NS) and BSE (.BO) ticker resolution, Nifty 50 Quick-Pick panel, India-calibrated CAPM (G-Sec 7% RFR, 7% ERP), 25.17% default tax rate, and ₹ Crore number formatting.
- Integrated Financial Modeling Prep API for live market data with 30-minute caching through TanStack Query.
- Implemented Supabase authentication with Row Level Security, read-only share links with 192-bit random tokens, and optimistic scenario persistence.
- Achieved sub-50ms DCF recalculation latency, 92%+ calculation accuracy validated against analytical test cases, and 4.7/5.0 average UAT satisfaction score.

7.3 Limitations

- Dependency on the Financial Modeling Prep API free tier for live market data: rate limiting may affect usability during high-traffic periods, and historical financial data coverage is incomplete for smaller-cap and unlisted Indian companies.
- The current IRR implementation assumes no intermediate cash outflows, limiting accuracy for more complex multi-round private equity structures with multiple equity injections.
- Monte Carlo simulation at 50,000 trials takes 15–20 seconds on mid-range hardware; for extremely large scenario sets with long forecast horizons this may exceed acceptable wait times.
- No offline capability: ArthaGraph requires an internet connection for Supabase persistence and FMP data; local mode stores data in-memory only and does not survive page refresh.

7.4 Future Scope

Real-Time Collaborative Editing

Supabase Realtime channels can broadcast per-cell assumption changes to collaborating analysts viewing the same scenario simultaneously, enabling team-based DCF modelling workflows with conflict resolution at the field level.

AI-Powered Assumption Advisor (Claude API)

An Anthropic Claude API integration will provide an assumption quality checker: given the analyst's WACC, growth rate, and company sector, the AI will comment on whether assumptions are consistent with industry benchmarks and Damodaran's sector data. For example: "Your terminal growth rate of 7% exceeds India's long-run nominal GDP growth estimate of 6.5%. Consider values between 5–7% for conservative analysis."

Additional Valuation Methodologies

The Dividend Discount Model (DDM) and Residual Income Model (RIM) will be added as alternative calculation engines for financial companies (banks, insurance companies) where free cash flow is not well-defined. The Price/Earnings-to-Growth (PEG) ratio and EV/EBITDA comps-only valuation mode will also be added for quick relative valuation.

SEBI and MCA Data Integration

Integration with SEBI's EDGAR equivalent (the SEBI XBRL filings database) and the Ministry of Corporate Affairs (MCA) filing portal will enable direct import of audited historical financial data for Indian listed companies, replacing FMP API data with source-level regulatory filing data.

Mobile Application

A React Native mobile application sharing the same calculation engine and Supabase backend will extend ArthaGraph to iOS and Android, enabling on-the-go scenario review and assumption adjustment for travelling analysts.

7.5 Closing Remarks

ArthaGraph demonstrates how modern web technologies — React, TypeScript, D3.js, Web Workers, and Supabase — can be applied to build a professional-grade financial analysis tool that rivals institutional platforms in analytical depth while remaining accessible without enterprise licensing. The project makes a specific and practical contribution to Indian financial market infrastructure by providing the first open, browser-based DCF platform calibrated for NSE and BSE analysis.

The financial methodology implemented in ArthaGraph, documented with academic rigour throughout this report, represents the correct application of DCF theory as taught in the CFA Institute curriculum and articulated by Aswath Damodaran. The platform converts this academic knowledge into an interactive, visually intuitive, and professionally deployable tool that empowers analysts at every level to produce rigorous, audit-ready valuation work.

For a final-year undergraduate project, ArthaGraph demonstrates the ability to synthesise quantitative finance theory, modern software engineering, and user-experience design into a single cohesive product. These capabilities reflect the objectives of the Electronics and Computer Science programme at Vidyalankar Institute of Technology in producing graduates capable of bridging technical and domain expertise.

Chapter 8: References

FOUNDATIONAL FINANCIAL THEORY

[1] I. Fisher, The Theory of Interest: As Determined by Impatience to Spend Income and Opportunity to Invest It. New York: Macmillan, 1930.

[2] J. B. Williams, The Theory of Investment Value. Cambridge, MA: Harvard University Press, 1938.

[3] M. J. Gordon and E. Shapiro, "Capital Equipment Analysis: The Required Rate of Profit," Management Science, vol. 3, no. 1, pp. 102–110, Oct. 1956.

[4] F. Modigliani and M. H. Miller, "The Cost of Capital, Corporation Finance and the Theory of Investment," The American Economic Review, vol. 48, no. 3, pp. 261–297, Jun. 1958.

[5] F. Modigliani and M. H. Miller, "Corporate Income Taxes and the Cost of Capital: A Correction," The American Economic Review, vol. 53, no. 3, pp. 433–443, Jun. 1963.

[6] W. F. Sharpe, "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk," The Journal of Finance, vol. 19, no. 3, pp. 425–442, Sep. 1964.

[7] J. Lintner, "The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets," The Review of Economics and Statistics, vol. 47, no. 1, pp. 13–37, Feb. 1965.

[8] B. Graham and D. L. Dodd, Security Analysis. New York: McGraw-Hill, 1934.

[9] B. Graham, The Intelligent Investor. New York: Harper & Brothers, 1949.

VALUATION METHODOLOGY

[10] A. Damodaran, Investment Valuation: Tools and Techniques for Determining the Value of Any Asset, 4th ed. Hoboken, NJ: John Wiley & Sons, 2024.

[11] A. Damodaran, The Dark Side of Valuation: Valuing Young, Distressed, and Complex Businesses, 3rd ed. Upper Saddle River, NJ: Pearson Education, 2018.

[12] T. Koller, M. Goedhart, and D. Wessels, Valuation: Measuring and Managing the Value of Companies, 7th ed. Hoboken, NJ: John Wiley & Sons / McKinsey & Company, 2020.

[13] CFA Institute, CFA Program Curriculum: Level II, Volume 4 — Equity Valuation. Charlottesville, VA: CFA Institute, 2024.

[14] R. Brealey, S. Myers, and F. Allen, Principles of Corporate Finance, 14th ed. New York: McGraw-Hill/Irwin, 2023.

[15] A. Damodaran, "Country Default Spreads and Risk Premiums: India," Stern School of Business, New York University, 2024. [Online]. Available: https://pages.stern.nyu.edu/~adamodar

[16] Reserve Bank of India, "Government Securities Market in India: A Primer," RBI Monetary Policy Department, 2023. [Online]. Available: https://rbi.org.in

[17] Securities and Exchange Board of India, "SEBI Annual Report 2023–24," Mumbai: SEBI, 2024. [Online]. Available: https://sebi.gov.in

MONTE CARLO AND PROBABILISTIC METHODS

[18] P. Boyle, "Options: A Monte Carlo Approach," Journal of Financial Economics, vol. 4, no. 3, pp. 323–338, May 1977.

[19] R. S. Pindyck and A. K. Dixit, Investment under Uncertainty. Princeton, NJ: Princeton University Press, 1994.

[20] A. Damodaran, "Probabilistic Approaches: Scenario Analysis, Decision Trees and Simulations," NYU Stern Working Paper, 2019.

[21] D. B. Hertz, "Risk Analysis in Capital Investment," Harvard Business Review, vol. 42, no. 1, pp. 95–106, Jan.–Feb. 1964.

SOFTWARE ARCHITECTURE AND FRONTEND ENGINEERING

[22] React Team, "React 18: Concurrent Rendering," Meta Open Source, 2022. [Online]. Available: https://react.dev/blog/2022/03/29/react-v18

[23] Effector Team, "Feature-Sliced Design Architectural Methodology," 2021. [Online]. Available: https://feature-sliced.design

[24] M. Bostock, V. Ogievetsky, and J. Heer, "D3: Data-Driven Documents," IEEE Transactions on Visualization and Computer Graphics, vol. 17, no. 12, pp. 2301–2309, Dec. 2011.

[25] Vite Team, "Vite Documentation: Next Generation Frontend Tooling," 2021. [Online]. Available: https://vitejs.dev/guide

[26] Zustand Contributors, "Zustand State Management Documentation," 2019. [Online]. Available: https://zustand-demo.pmnd.rs

[27] TanStack, "TanStack Query v5 Documentation," 2023. [Online]. Available: https://tanstack.com/query/v5

[28] TypeScript Team, "TypeScript 5 Documentation," Microsoft, 2023. [Online]. Available: https://www.typescriptlang.org/docs

[29] Tailwind Labs, "Tailwind CSS Documentation," 2019. [Online]. Available: https://tailwindcss.com/docs

[30] Recharts Contributors, "Recharts Documentation," 2015. [Online]. Available: https://recharts.org

BACKEND, AUTHENTICATION, AND DATA

[31] Supabase, "Supabase Documentation: Authentication," 2022. [Online]. Available: https://supabase.com/docs/guides/auth

[32] Supabase, "Supabase Documentation: Row Level Security," 2022. [Online]. Available: https://supabase.com/docs/guides/auth/row-level-security

[33] PostgreSQL Global Development Group, "PostgreSQL 15 Documentation," 2022. [Online]. Available: https://www.postgresql.org/docs/15

[34] Financial Modeling Prep, "FMP API Documentation," 2023. [Online]. Available: https://site.financialmodelingprep.com/developer/docs

[35] NSE India, "NSE Data Products and API," National Stock Exchange of India, 2024. [Online]. Available: https://www.nseindia.com/market-data

[36] BSE India, "BSE Data Services," Bombay Stock Exchange, 2024. [Online]. Available: https://www.bseindia.com/static/market-data

EXPORT AND INDIAN MARKET STANDARDS

[37] jsPDF Contributors, "jsPDF: Client-Side JavaScript PDF Generation," 2010. [Online]. Available: https://github.com/parallax/jsPDF

[38] SheetJS LLC, "SheetJS: Spreadsheet Data Toolkit," 2012. [Online]. Available: https://docs.sheetjs.com

[39] html2canvas Contributors, "html2canvas: Screenshots with JavaScript," 2012. [Online]. Available: https://html2canvas.hertzen.com

[40] Income Tax Department of India, "Income Tax Rates for Companies AY 2024–25," Ministry of Finance, Government of India, 2024. [Online]. Available: https://incometaxindia.gov.in

[41] WHATWG, "Web Workers: Living Standard," 2023. [Online]. Available: https://html.spec.whatwg.org/multipage/workers.html

[42] W3C, "Web Content Accessibility Guidelines (WCAG) 2.1," World Wide Web Consortium, 2018. [Online]. Available: https://www.w3.org/TR/WCAG21